

1 ECO 310: Precept 2 - Optimization

1.1 Intuitive Method and "Non-Standard" Budget Constraints

Consider a consumer who cares about consumption today (C_1) and consumption tomorrow (C_2), with preferences $u(C_1, C_2) = C_1 + 0.5C_2$. (So consumption today and consumption tomorrow are perfect substitutes, but tomorrow is only half as important as today). He has \$100 available, but can invest whatever he doesn't spend today at interest rate r , meaning that \$ w invested will be worth \$ $w(1 + r)$ tomorrow. Assume a nonnegativity restriction on C_1 and C_2

Using the intuitive method, find his optimal consumption levels.

Solution: if he spends $C_1 \leq 100$ today, he will invest the remaining $\$(100 - C_1)$ at interest rate r , and thus will be able to spend $(1 + r)(100 - C_1)$ tomorrow. So his budget constraint is $C_2 \leq (1 + r)(100 - C_1)$. To apply the intuitive method, let's rearrange this to put consumption goods on the LHS and resources on the RHS:

$$(1 + r)C_1 + C_2 \leq 100(1 + r) \quad (*)$$

Now, for this example, $\frac{MU_1}{MU_2} = \frac{1}{0.5} = 2$ (consumption today is twice as valuable as consumption tomorrow, but they are perfect substitutes). But we can look at the coefficients on C_1 and C_2 in the budget line to get their prices, namely $\frac{p_1}{p_2} = \frac{1+r}{1} = 1 + r$. So:

- if an interior solution is optimal, it solves $\frac{MU_1}{MU_2} = \frac{p_1}{p_2}$ (equal bang-per-buck across goods), so $2 = 1 + r \Leftrightarrow r = 1$. Interpretation: if the return rate happens to be exactly 1 (100%), then ANY allocation of funds between consumption today/tomorrow leads to equal bang-per-buck across goods, and thus is optimal. So any consumption level between 0 and 100 today, with the rest invested and saved for tomorrow, is optimal.
- if the return rate exceeds 1, then $\frac{MU_1}{MU_2} = 2$ while $\frac{p_1}{p_2} = 1 + r > 2$. So no matter what the consumer does, $\frac{MU_1}{MU_2} < \frac{p_1}{p_2} \Leftrightarrow \frac{MU_1}{p_1} < \frac{MU_2}{p_2}$, i.e. good 1 (today's consumption) has less bang-per-buck than good 2 (tomorrow's consumption). In this case, a corner solution is optimal: the consumer should put all his money where it has the most bang-per-buck, namely consume nothing today, $C_1 = 0$, and instead invest the entire \$100 and consume the resulting $\$100(1 + r)$ tomorrow.
- and finally if the return rate is smaller than 1, then $\frac{MU_1}{MU_2} = 2$ while $\frac{p_1}{p_2} = 1 + r < 2$. So no matter what the consumer does, $\frac{MU_1}{MU_2} > \frac{p_1}{p_2} \Leftrightarrow \frac{MU_1}{p_1} > \frac{MU_2}{p_2}$, i.e. consumption today (good 1) has more bang-per-buck than consumption tomorrow (good 2). And so a corner solution is optimal where the consumer consumes the entire \$100 today and saves nothing for tomorrow.

1.2 Lagrangians

Now use a Lagrangian to solve the problem above, in the case where $r < 1$. Use the solution above to guess which constraints will bind.

Solution: first form the Lagrangian: write out what you want to maximize, add any expressions that must be ≥ 0 (each multiplied by its own Lagrange multiplier), or equivalently subtract any expressions that must be ≤ 0 (each multiplied by a multiplier). Written in $\geq$ form here, our constraints are $C_1 \geq 0$, $C_2 \geq 0$, and $100(1 + r) - (1 + r)C_1 - C_2 \geq 0$. So:

$$\mathcal{L} = C_1 + \frac{1}{2}C_2 + \lambda_1 C_1 + \lambda_2 C_2 + \lambda_3 [100(1 + r) - (1 + r)C_1 - C_2]$$

Now,

- for each of the three constraints, either it binds (holds with equality and has a positive Lagrange multiplier, meaning does impact the solution so you optimally *just* satisfy it) or doesn't bind (holds and has a zero Lagrange multiplier, meaning "you can ignore it, by setting $\lambda = 0$ (which wipes it out of the problem), and will happen to satisfy it anyway")
- We know that the budget constraint always binds (you always wish you had more money to spend), so this gives us one equation:

$$\lambda_3 > 0 \text{ and } (1+r)C_1 + C_2 = 100(1+r) \quad (5)$$

- we saw above that for the specified range $r < 1$, the consumer spends all income on year-1 consumption, since it is a better use of money. This suggests that if they *could*, they'd choose $C_2 < 0$ to finance even more C_1 , whereas we see they want positive C_1 anyway. So our guess is that the constraint $C_2 \geq 0$ does bind (they wish they could violate it) but $C_1 \geq 0$ does not (this is what the consumer wants anyway). So let's look for a solution where this happens: non-binding constraint $C_1 \geq 0$ means

$$\lambda_1 = 0 \text{ (and anyway choose } C_1 \geq 0) \quad (4)$$

and binding constraint on $C_2 \geq 0$ means

$$\lambda_2 > 0 \text{ and } C_2 = 0 \quad (3)$$

(it restricts you by a positive amount, captured by $\lambda_2 > 0$, and so you optimally respond by just barely satisfying the rule you'd prefer to ignore: barely satisfying $C_2 \geq 0$ means setting C_2 exactly equal to zero). Finally, we have the usual FOC's on the choice variables C_1 and C_2 :

$$0 = \frac{\partial \mathcal{L}}{\partial C_1} = 1 + \lambda_1 - \lambda_3(1+r) \quad (1)$$

$$0 = \frac{\partial \mathcal{L}}{\partial C_2} = \frac{1}{2} + \lambda_2 - \lambda_3 \quad (2)$$

Now, we want to solve (1)-(5) for C_1, C_2 , and if desired, the λ 's. We already have $\lambda_1 = 0$ (from (4)) and $C_2 = 0$ (from (3)). Plugging $C_2 = 0$ into (5) gives

$$(1+r)C_1 + 0 - 100(1+r) = 0 \Rightarrow C_1 = 100$$

Plugging $\lambda_1 = 0$ into (1) gives $\lambda_3 = \frac{1}{1+r}$ (positive as we assumed in (5)), and plugging this into (2) gives $\lambda_2 = \frac{1}{1+r} - \frac{1}{2}$ (positive, as we assumed in (3), provided that $r < 1$, which was indeed the premise of the question).

So we're done: if $r < 1$, the solution is $C_1 = 100, C_2 = 0, \lambda_3 = \frac{1}{1+r}, \lambda_2 = \frac{1}{1+r} - \frac{1}{2}, \lambda_1 = 0$, which solves all of (1)-(5).